For the exclusive use of C. Su, 2026.

IN2027

The INSEAD Blue Ocean Strategy Institute

Case Study

NVIDIA's Future Strategy:

Can It Sustain Its Blue Ocean?

01/2025-6964

This case study was written by Zunaira Munir, Institute Executive Fellow of the INSEAD Blue Ocean Strategy Institute, under the supervision of W. Chan Kim and Renée Mauborgne, Professors of Strategy at INSEAD. It is intended to be used as a basis for class discussion rather than to illustrate either effective or ineffective handling of an administrative situation.

To access INSEAD teaching materials, go to https://publishing.insead.edu/.

Copyright © 2025 INSEAD. All rights reserved.

NO PART OF THIS PUBLICATION MAY BE COPIED, STORED, TRANSMITTED, TRANSLATED, REPRODUCED OR DISTRIBUTED IN ANY FORM, MEDIUM, APPLICATION, OR SYSTEM WITHOUT THE PERMISSION OF THE COPYRIGHT OWNER. SUCH RESTRICTION SHALL ALSO COVER AI SYSTEMS AND AI MODELS.

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

On November 30, 2022, OpenAI released ChatGPT to the world. The announcement read, "We've trained a model called ChatGPT, which interacts in a conversational way. The dialogue format allows ChatGPT to answer follow-up questions, admit mistakes, challenge incorrect premises, and reject inappropriate requests."[1]

Generative artificial intelligence (AI) was no longer a futuristic concept. The AI chatbot spoke your language, understood the context, communicated with you with human-like finesse, and was within everybody's reach. The world took notice. Within a month, over a hundred million users rushed to chatgpt.com to try it out.

The release marked a significant shift in AI. This wasn't just about cool tech tricks. AI was now seen as a foundational technology with almost unlimited applications, opening new ways for businesses to automate tasks, streamline processes, and innovate faster at lower cost. Practically every industry started looking at new AI functionality to enhance productivity, efficiency, and creativity. Venture capital financing poured millions into the many startups developing a new generation of generative AI-powered applications. Recognizing the immense potential, tech giants raced to establish their dominance in the field. The *New York Times* reported that once ChatGPT was unleashed, "the instinct to be first or biggest or richest—or all three—took over. The leaders of Silicon Valley's biggest companies set a new course and pulled their employees along with them. Over 12 months, Silicon Valley was transformed. Turning artificial intelligence into actual products that individuals and companies could use became the priority."[2] Even non-tech companies recognized the value of generative AI and sought to incorporate it into their internal operations. Goldman Sachs estimates that applications of this technology could raise global GNP by $7 trillion during the next decade.[3]

At the center of this AI surge was NVIDIA.

The company's chips became must-haves for anyone wanting to be a player in the AI space. The result was a sudden boom in sales and skyrocketing stock prices. NVIDIA's stock market rise has been among the fastest in history. In 2022, the company's market valuation was a little over $400 billion. Within the span of a year, from 2023 to 2024, it went from $1 trillion to over $3 trillion.[4] It's the sort of stock market price gains that can create generational wealth for investors. Indeed, a $10,000 investment in NVIDIA 10 years ago was worth more than $2 million in 2024.[5]

---

1 OpenAI. (2022, November 30). *Introducing ChatGPT.* https://openai.com/index/chatgpt/

2 Weise, K., Metz, C., Grant, N., & Isaac, M. (2023, December 5). Inside the A.I. arms race that changed Silicon Valley forever. *The New York Times*. https://www.nytimes.com/2023/12/05/technology/ai-chatgpt-google-meta.html

3 Goldman Sachs. (2023, April 5). *Generative AI could raise global GDP by 7%.* https://www.goldmansachs.com/insights/articles/generative-ai-could-raise-global-gdp-by-7-percent

4 Mickle, T., & Rennison, J. (2024, June 18). Nvidia becomes most valuable public company, topping Microsoft. *The New York Times*. https://www.nytimes.com/2024/06/18/technology/nvidia-most-valuable-company.html

5 Pope, J. (2024, July 4). Could investing $10,000 in Nvidia make you a millionaire? *Nasdaq.* https://www.nasdaq.com/articles/could-investing-10000-NVIDIA-make-you-millionaire

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

But is NVIDIA on a sustainable, high-growth path that the market has confidence in? Or are challenges emerging on the horizon that could get in NVIDIA's way? Can NVIDIA continue to swim in a blue ocean devoid of any competition, or are sharks lurking nearby that are likely to soon turn the ocean bloody red? How is the strategic landscape likely to change, and what does NVIDIA need to do to stay ahead of its competitors? Important questions loom for the company.

The Embarrassingly Parallel AI Problem

In computer science, an embarrassingly parallel workload is one that can be split into a number of smaller similar tasks with little or no effort to be processed simultaneously by multiple processing units. Deep learning on artificial neural networks (ANN) that enables generative AI is one such embarrassingly parallel problem (See Appendix 1, Background Note on Generative AI, for more details). The deep learning process involves training a neural network on a large dataset so that the network adjusts its internal parameters to minimize the error between its predicted outputs and the actual target outputs. This iterative process, called backpropagation, is computationally intensive and allows the network to learn intricate relationships and make accurate predictions on new, unseen data. The problem is inherently a parallel one. By dividing the computational workload and distributing the computations across multiple processing units, the entire deep learning process can be done much faster (or accelerated) using a parallel processor like the graphics processing unit (GPU), which NVIDIA pioneered in 1999, starting a new wave of what became known as accelerated computing.

Handling and processing such large datasets and iterative optimization processes requires powerful processors, robust storage, and rapid data retrieval systems. Moreover, these processes can be extremely energy intensive, producing a lot of heat and necessitating energy-efficient cooling systems to minimize power usage. In short, the development and execution of AI applications at scale requires specialized data centers with extensive computing resources and infrastructure optimized for running AI workloads.

With the worldwide realization of how useful AI can be to businesses and their bottom lines, demand for NVIDIA GPUs has soared. In 2023, NVIDIA shipped an estimated 3.76 million GPUs for data centers, over 1 million more than the 2.64 million units it shipped in 2022.[6] In contrast, AMD and Intel collectively shipped 710,000 AI accelerators, making NVIDIA's share of the GPU market almost 98%. Data center business brought NVIDIA $47.5 billion in revenues for 2023, a 217% rise over 2022. NVIDIA's total revenue was up 126% to a record $60.9 billion, with a gross margin of 72.7%. The explosive growth of NVIDIA's data center business has made everything else the company does feel relatively incremental. Gaming and automotive combined typically

6 Trueman, C. (2024, June 12). Nvidia data center GPU shipments totaled 3.76m in 2023, equating to a 98% market share—Report. *Data Center Dynamics*. https://www.datacenterdynamics.com/en/news/nvidia-gpu-shipments-totaled-376m-in-2023-equating-to-a-98-market-share-report/

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

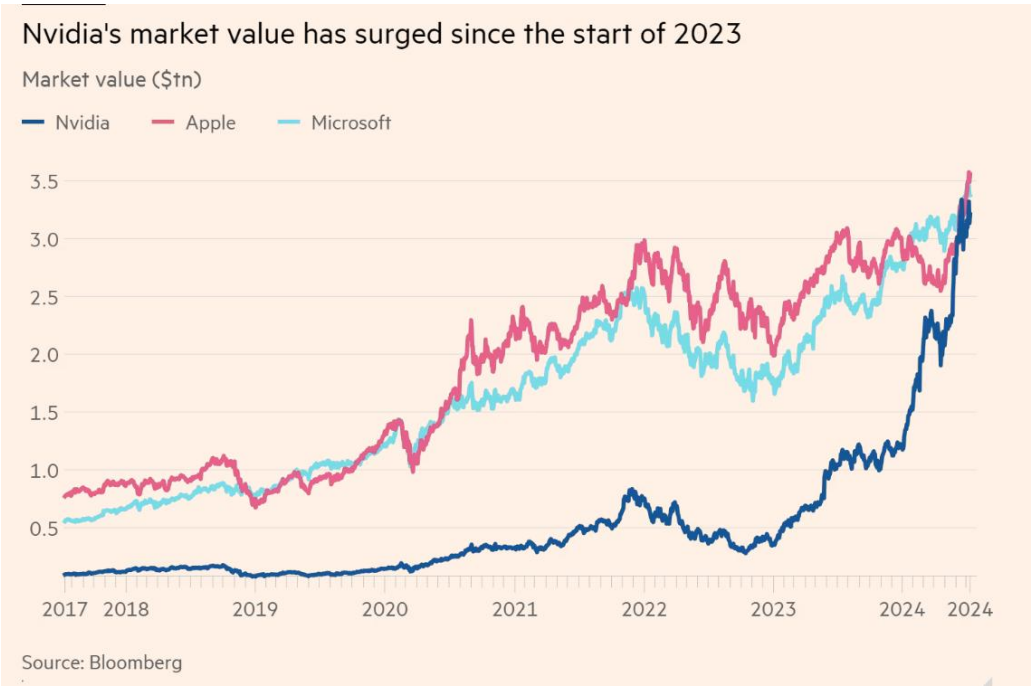
For the exclusive use of C. Su, 2026.

accounted for more than half of NVIDIA's annual revenue; analysts expect that contribution to be just around 19% in 2024.

A 2024 TechInsights report indicates that NVIDIA is emerging as a clear winner. There isn't enough AI hardware to match the rapid progress in AI software, and there are only a few alternatives to NVIDIA's hardware on the market. These include Google's Tensor Processing Units (TPUs), AMD's Instinct AI accelerators, and Intel's Gaudi AI accelerators and central processing units (CPUs) that come with Neural Processing Units (NPUs).[7] NVIDIA's main rival, AMD, has projected revenue from its Instinct line of AI chips in 2024 at $4.5 billion. In contrast, NVIDIA expects $76 billion and $91 billion in revenue from data centers in 2024 and 2025, respectively.

Figure 1 below shows NVIDIA's unprecedented stock market rise, which made it the most valuable company in the world in 2024, bypassing Microsoft and Apple. Figure 2 shows FY 2024 financials (year ending Jan 2024), with a summary and breakdown of revenue per product line. Figure 3 shows how revenues per product line have changed over the past five years.

Figure 1: NVIDIA's unprecedented stock market rise



Source: Bloomberg.

7 Shah, A. (2024, June 10). Nvidia shipped 3.76 million data-center GPUs in 2023, according to study. *HPCwire*. https://www.hpcwire.com/2024/06/10/nvidia-shipped-3-76-million-data-center-gpus-in-2023-according-to-study/

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

The INSEAD Blue Ocean Strategy Institute

Figure 2: NVIDIA Fiscal Year 2024 summary

Fiscal Year 2024 Summary

	Year Ended		
	Jan 28, 2024	Jan 29, 2023	Change
	($ in millions, except per share data)		
Revenue	$ 60,922	$ 26,974	Up 126%
Gross margin	72.7 %	56.9 %	Up 15.8 pts
Operating expenses	$ 11,329	$ 11,132	Up 2%
Operating income	$ 32,972	$ 4,224	Up 681%
Net income	$ 29,760	$ 4,368	Up 581%
Net income per diluted share	$ 11.93	$ 1.74	Up 586%

	Year Ended		
	Jan 28, 2024	Jan 29, 2023	Jan 30, 2022
Revenue:	*(In millions)*		
Data Center	$ 47,525	$ 15,005	$ 10,613
Gaming	10,447	9,067	12,462
Professional Visualization	1,553	1,544	2,111
Automotive	1,091	903	566
OEM and Other	306	455	1,162
Total revenue	$ 60,922	$ 26,974	$ 26,914

Source: NVIDIA 2024 Annual Report.

Figure 3: NVIDIA's breakdown of revenues per product line

Product Line	2024	2023	2022	2021	2020	2019
Data Center Processors for Analytics and AI	78.0%	55.6%	39.4%	40.2%	27.3%	25.0%
GPUs for Computers	17.1%	33.6%	46.3%	46.5%	50.5%	53.3%
GPUs for 3D Visualization	2.6%	5.7%	7.8%	6.3%	11.1%	9.6%
GPUs for Automotive	1.8%	3.3%	2.1%	3.2%	6.4%	5.5%
GPUs for Cryptocurrency Mining	0.0%	0.0%	2.0%	3.8%	4.6%	6.5%
Other	0.5%	1.7%	2.3%	0.0%	0.0%	0.0%

Source: NVIDIA 2024 Annual Report

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

Analysts say that the fear of missing out (FOMO) is high, and almost everyone seems to be hopping on the AI wave. Big tech companies like Alphabet, Microsoft, Amazon, Meta Platforms, and others have been committing billions of dollars to NVIDIA to get their hands on the next batch of chips. As a result, customers want more of those chips than NVIDIA can produce, making it effectively backlogged for orders through 2025. While the most prominent tech firms have been able to get their hands on NVIDIA GPUs more easily because of their size, deep pockets, and market positions, startups are left scrambling. In the meantime, antitrust authorities are becoming concerned about NVIDIA's power to determine how scarce but essential technology is being allocated.

On Google's recent earnings call, Alphabet CEO Sundar Pichai summarized why big tech companies continue to purchase every chip they can from NVIDIA: "When we go through a curve like this, the risk of under-investing is dramatically greater than the risk of over-investing for us."[8]

As the *New York Times* reported in August 2024, "Customers will wait 18 months to buy an NVIDIA system rather than buy an available, off-the-shelf chip from either a startup or another competitor." Elon Musk tweeted, "GPUs are harder to get than drugs." In a cultural twist to this tech dilemma, a YouTube parody made to the tune of Billy Joel's "We Didn't Start the Fire" went viral, expressing the industry's frustration in securing NVIDIA's GPUs: "GPUs are fire, we can never find 'em, but we wanna buy 'em."[9]

NVIDIA's Walled Garden

Analysts say that NVIDIA has built a "walled garden" that protects its dominance in AI.[10] In the context of technology and computing, a walled garden refers to a closed ecosystem in which the ecosystem operator controls all the operations and users can only access the resources that the operator allows. NVIDIA's walled garden, they say, is its hardware-software-developer ecosystem. As the *New York Times* reported,

Over more than 10 years, NVIDIA has built a nearly impregnable lead in producing chips that can perform complex AI tasks like image, facial, and speech recognition, as well as generating text for chatbots like ChatGPT. The onetime industry upstart achieved that dominance by recognizing

8 Kindig, B. (2024, August 8). Big Tech battles on AI, here's the winner. *Forbes*. https://www.forbes.com/sites/bethkindig/2024/08/08/microsoft-leads-big-tech-in-ai-monetization-amazon-a-close-second/

9 "Weird A.I." Yankochip. (2023, July 24). *The GPU song (GPUs are fire)* [Video]. YouTube. https://www.youtube.com/watch?v=xG6VPA_rGM0

10 Mims, C. (2024, August 10). Why every Big Tech company has failed to dethrone Nvidia as king of AI. *The Wall Street Journal*. https://www.wsj.com/tech/ai/ai-nvidia-apple-amd-jensen-huang-software-bb581f5a

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

the AI trend early, tailoring its chips to those tasks, and then developing key pieces of software that aid in AI development.[11]

Since the 1999 launch of its first GPU, the GeForce 256, NVIDIA has consistently pushed the boundaries of computing and set new hardware benchmarks by introducing a series of cutting-edge architectures, often named after prominent world scientists. Each release represents a significant leap forward in technology and delivers unprecedented value to customers.[12] In 2007, NVIDIA's Tesla architecture laid the foundation for accelerated computing by making its GPUs programmable. V100, based on NVIDIA's Volta architecture, was designed specifically for deep learning and high-performance computing applications. In 2018, NVIDIA's CEO Huang commented that GPUs were growing 25 times faster than in 2013, doubling their performance every year. A100, based on NVIDIA's Ampere architecture and introduced in 2020, has become the gold standard for data centers. H100, based on its Hopper architecture and introduced in 2023, delivers nine times faster AI training and 30 times speedier AI inference compared with A100.

The performance of GPUs is measured in floating point operations per second (FLOPS), the number of discrete arithmetic calculations a chip can perform in one second. As the processing power of NVIDIA's leading chips grew exponentially, the unit for expressing FLOPS in technical specifications evolved from MegaFLOPS (1,000,000 FLOPS) to GigaFLOPS (1,000,000,000) to TeraFLOPS (1,000,000,000,000) to PetaFLOPS (one quadrillion or 10^{15}) and now to ExaFLOPS (one quintillion or 10^{18} FLOPS). NVIDIA's latest offering for the data center market, the NVIDIA GB200 NVL72, delivers 1.4 ExaFLOPS of AI performance while reducing cost and energy consumption by up to 25 times as compared with the previously available options. It is a data center rack-scale product that acts as a single GPU. It comprises 36 Grace Blackwell Superchips, each with 72 Blackwell GPUs, 36 Grace CPUs, and three BlueField DPUs, all interconnected by fifth-gen NVLink for hyperscale AI cloud delivery.[13]

As important as the raw computational power measured in FLOPS is to AI users, the most critical hardware consideration for building out AI infrastructure is FLOPS per dollar, a measure of the cost-effectiveness of the chip. Each new generation of NVIDIA's GPUs offered higher performance at lower FLOPS per dollar. In 2023, an A100 cost around $10,000, while the newer H100s were priced at $40,000. At these prices, the A100 came in at around 0.00195 TFLOPS/dollar compared with the H100's 0.00168 TFLOPS/dollar. Analysts say, "Even though the cost of the system is going up, the performance per dollar is getting better with every

---

11 Clark, D. (2023, August 21). How Nvidia built a competitive moat around A.I. chips. *The New York Times*. https://www.nytimes.com/2023/08/21/technology/nvidia-ai-chips-gpu.html

12 NVIDIA introduced cutting-edge architectures such as Fahrenheit (1998), Celsius (1999), Kelvin (2001), Rankine (2003), Curie (2004), Tesla (2006), Fermi (2010), Kepler (2012), Maxwell (2014), Pascal (2016), Volta (2017), Turing (2018), Ampere (2020), Hopper (2022), Ada (2022), Blackwell (2024), and Rubin (coming in 2026).

13 In addition to GPUs, over the years, NVIDIA has also developed cutting-edge central processing units (CPUs), data processing units (DPUs), and also system-on-a-chip, all designed for specific functionalities.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

generation, and that's how they're able to keep customers buying."[14] Based on the improvements from the V100 (with which GPT-3 was trained), H100 would reduce in-house training costs from $744,000 to an estimated $312,000.

NVIDIA's walled garden is made equally of software and hardware. The *Wall Street Journal* wrote that the key to understanding NVIDIA's walled garden is their software platform, called Compute Unified Device Architecture (CUDA). "It is what keeps customers in and competitors out."[15] NVIDIA's GPUs were initially designed to accelerate 3D graphics processing. CUDA, first released in 2007, enabled those graphics chips to be programmed and optimized for all kinds of other computing tasks that could benefit from the parallel processing capabilities of NVIDIA GPUs. And the release of CUDA was just the beginning. Year after year, NVIDIA responded to the needs of software developers by pumping out specialized libraries of code, allowing a massive array of tasks to be performed on its GPUs at speeds that were impossible with conventional, general-purpose processors or CPUs made by Intel and AMD. This meant that applications running on NVIDIA GPUs using CUDA could achieve significant performance gains compared with CPUs or any other accelerators not supported by CUDA.

While NVIDIA contributes to some open standards, it has kept its core CUDA platform proprietary. This means that CUDA is specifically designed to work with NVIDIA GPUs and can't be used directly with GPUs from other manufacturers like AMD or Intel. Developers in the NVIDIA ecosystem don't need to build a product from the ground up. Instead, they can take advantage of curated tools, libraries, and compilers that NVIDIA provides, reducing their work to a small part of what would otherwise be a cumbersome technical process. However, to take advantage of these features, they must use NVIDIA GPUs, not any other. This ecosystem effectively ties developers to NVIDIA hardware and creates a virtuous cycle that reinforces NVIDIA's dominance in AI chips. The performance benefits offered by CUDA attract developers who contribute to the large body of knowledge and compatible resources that benefit new and existing CUDA users. This drives more adoption by attracting more developers, creating substantial switching costs and a powerful moat around NVIDIA's core AI offerings. Replicating this integrated ecosystem presents a daunting challenge to rivals, analysts say.

An entire cohort of developers have been working with CUDA for years, and there is little reason for them to consider using another software platform. CUDA now includes more than 300 code libraries and 600 AI models and supports 3,700 GPU-accelerated applications used by more than five million developers at roughly 40,000 companies. The NVIDIA platform is also available through every major cloud provider and computer maker. This breadth and accessibility effectively make NVIDIA the go-to platform for AI developers. When a rival chip is introduced, it

14 Mims, C. (2024, August 10). Why every Big Tech company has failed to dethrone Nvidia as king of AI. *The Wall Street Journal*. https://www.wsj.com/tech/ai/ai-nvidia-apple-amd-jensen-huang-software-bb581f5a

15 Mickle, T., & Rennison, J. (2024, June 18). Nvidia becomes most valuable public company, topping Microsoft. *The New York Times*. https://www.nytimes.com/2024/06/18/technology/nvidia-most-valuable-company.html

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

is not just up against NVIDIA chips. It is up against systems that NVIDIA's developers have been using for more than 15 years to write mountains of code. That software is very difficult to shift to a competitor's system.

The *Wall Street Journal* writes, "The importance of NVIDIA's software platforms also explains why for years NVIDIA has had more software engineers than hardware engineers on its staff…. It also explains why, in the longer term, the battle over territory NVIDIA now dominates is likely to focus on the company's coding prowess, not just its circuitry design—and why its rivals are racing to develop software that can circumvent NVIDIA's protective wall."[16]

NVIDIA's walled garden has given it significant pricing power and allows it to operate at stunningly high 73% gross margins. Rival chipmakers Intel and AMD, in comparison, have reported 41% and 47% gross margins, respectively. Analysts say that NVIDIA has been creating new markets, and that puts competitors in a very tough situation because by the time they've caught up, NVIDIA is on to the next new thing.[17]

The Race for Next-Gen AI Data Centers

More often than not, the data center of the early '90s was a converted utility room replete with computing, storage, and networking equipment to run and deliver custom applications and store and manage proprietary data for large companies. In the mid-2000s, a shift to data centralization and distributed computing occurred, moving computation and data storage from servers inside offices to large-scale data centers worldwide. When companies like Amazon and Microsoft began to build massive data center facilities and branched out to offer cloud services to other companies (e.g., Amazon Web Services in 2006), hyperscale data centers began to emerge, generally in areas with abundant energy reserves and good fiber-optic connectivity. These cloud service providers housed virtualized IT infrastructure for the shared use of multiple companies and started offering infrastructure as a service (IaaS), platform as a service (PaaS), and software as a service (SaaS).

Data centers are expensive to build and operate. Still, an organization that expected to use substantial computing power at times found it more cost-effective to build out a private data center than to rent computing power from a cloud provider. Some companies took a hybrid approach, using a cloud provider for lower-cost functions and doing more computationally intensive back-end operations on company-owned and -operated hardware. Since the 2010s, colocation data centers (colos) have been filling the gap between in-house build-outs of on-premise computing infrastructure and the public cloud offering data-center services. In 2014,

16 Mims, C. (2024, August 10). Why every Big Tech company has failed to dethrone Nvidia as king of AI. *The Wall Street Journal*. https://www.wsj.com/tech/ai/ai-nvidia-apple-amd-jensen-huang-software-bb581f5a

17 Howley, D. (2024, January 22). Nvidia's rivals are circling, but they're still years from catching up. *Yahoo Finance*. https://finance.yahoo.com/news/nvidias-rivals-are-circling-but-theyre-still-years-from-catching-up-211621835.html

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

80% of data centers were private enterprises, and 20% were colos or cloud. By 2023, enterprise data centers' share had fallen to 30%.

In recent years, hyperscale cloud data centers have become the backbone of the digital economy and play a critical role in enabling the internet and helping people stay connected and productive. International Data Center (IDC) Corporation defines a hyperscale data center as one containing at least 5,000 servers, occupying at least 10,000 square feet of physical space, and drawing over 100 megawatts (MW) of energy. The term "hyperscale" refers to data center size as well as its ability to scale up capacity in response to demand. Amazon Web Services (AWS), Google Cloud Platform (GCP), and Microsoft Azure are the leading hyperscale data center providers, with AWS commanding 32%, Microsoft Azure 23%, and Google Cloud Platform about 10% of the cloud services market. Other hyperscale data centers are operated by Alibaba, Apple, Meta Platforms, IBM Cloud, and Oracle Cloud Infrastructure (OCI).

At the beginning of 2024, there were approximately 10,978 data center locations worldwide, including more than a thousand hyperscale data centers.[18] The United States accounts for 51% of worldwide capacity, with Europe and China each accounting for about a third of the balance. Data center capacity measured as MW of IT load has more than doubled in the past four years, and the average data center size worldwide is 100,000 square feet (about 1.74 football fields). The complexity of building a hyperscale data center would be comparable to building a Boeing Dreamliner, analysts say, often costing hundreds of millions of dollars. Electricity is a significant cost to operate a data center, as data centers require large amounts of reliable power and extensive cooling equipment as they generate a substantial amount of heat. According to one estimate, the average yearly cost to operate a large data center can range from $10 million to $25 million. Data centers' energy use breakdown is generally 50% to run servers, 11% to run storage devices, 2% to run networks, and 37% to run cooling and energy provision systems.

Data centers are uniquely positioned to benefit from and provide for AI's increased computational needs. Analysts are expecting 120 to 130 additional higher-capacity hyperscale data centers to come online every year and data center capacity to double every four years.[19] What is particularly changed in the emerging AI era is the scale of computational demands required to train new models, which can generate very expensive cloud bills for companies that don't own their own computing resources.

The AI data center market is massive, and NVIDIA is well positioned to benefit from it, given the technological head start it has had in the AI space. NVIDIA has set its eyes on modernizing the world's trillion-dollar data center infrastructure. CEO Huang says, "Every data center in the world needs to be accelerated to reclaim power, achieve sustainability, and realize net-zero

18 Minnix, J. *255 data center stats (March 2026).* Brightlio. Retrieved July 15, 2026, from https://brightlio.com/data-center-stats/

19 Yadav, N. (2024, April 22). Hyperscale data center capacity to double every four years—Report. *Data Center Dynamics*. https://www.datacenterdynamics.com/en/news/hyperscale-data-center-capacity-to-double-every-four-years-report/

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

emissions."[20] According to NVIDIA, two of its HGX systems with H100 GPUs that together cost $500K can replace 1,000 nodes of CPU servers that cost $10 million for AI workloads. "Accelerated data centers could save an incredible 19 terawatt-hours of electricity annually if run on GPU and DPU accelerators vs CPUs. That's about the same energy as a year's worth of trips by 2.9 million passenger cars," NVIDIA's 2024 Annual Report states.[21]

But other players besides NVIDIA also have their eyes on the evolving data center market. The race for dominance in the next-generation AI data centers is heating up, and all major technology companies are investing heavily in sharing a claim in the new digital infrastructure. Some say the real battle is not for market share but for shaping the fabric of the new tech landscape. Hyperscalers say their colossal spending on digital infrastructure is an essential down payment on the game-changing AI technology, and companies have been increasing R&D spending. However, a lot of that R&D money is being routed back to NVIDIA as firms spend millions of dollars purchasing GPUs. What's cost to other tech giants is revenue to NVIDIA.

NVIDIA's Direct Competitors: AI Chip Manufacturers

Intel

Once considered the world's most advanced chip maker, Intel is one of the few integrated device manufacturers (IDMs) in the world that designs, fabricates, packages, and markets its own semiconductors. NVIDIA once faced an existential threat for its graphics cards when Intel started integrating graphics acceleration within the CPU chips they sold for personal computers (PCs). Intel had about 10 times NVIDIA's annual revenue then. NVIDIA surpassed Intel in total annual revenue in 2023 and is expected to close its fiscal year ending in January of 2025 with revenue of $91 billion—15% more than Intel's all-time annual record.

During the 90s, the combination of Intel x86 CPUs and OS software offered unmatched power to enterprises. Until recently, it remained the hardware provider of choice for data centers. Over the last few years, however, Intel has struggled to maintain its leadership position and has witnessed a drop in market share across the globe. It was slow to master the latest fabrication processes and has since been eclipsed by pure-play foundries like TSMC and Samsung that fabricate chips for the likes of NVIDIA and AMD. (See Appendix 2, Background Note on the Global Semiconductor Industry.)

Intel bought the Israeli AI startup Habana Labs in 2019 for about $2 billion and started producing its AI chips. It recently announced the third version of its AI accelerator, Gaudi 3, describing it as a more cost-effective alternative to NVIDIA's H100. With $8.5 billion in grants from the US

20 NVIDIA Corporation. (2024). *2024 NVIDIA Corporation annual review: Notice of annual meeting, proxy statement, Form 10-K.* https://s201.q4cdn.com/141608511/files/doc_financials/2024/ar/NVIDIA-2024-Annual-Report.pdf

21 Ibid.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

Government, Intel has five new fabrication plants in the works. It has recently launched Internet Foundry Services (IFS) and plans to offer its fabrication services to other chip makers. Opening its factories to other chip companies is no simple process, especially because many of those potential customers make chips that compete with Intel's. Bank of America analysts estimate that Intel will have less than 1% of the AI chip market in 2024.

Advanced Micro Devices (AMD)

AMD is seen as the closest rival to NVIDIA. The company is a longtime competitor, with its own line of AI processors and deep relationships with big data center operators hungry for computing power. AMD started designing and fabricating chips in 1969, a time when chip manufacturers were making the legendary boast, "Real men have fabs." It later spun off its fabrication business and became fabless in 2009. Like Intel, AMD draws much of its business from the mature PC market with its CPUs. In 2006, AMD acquired the graphics card company ATI, a major NVIDIA competitor at the time, and entered the accelerated computing market in 2009 with its combined CPU/GPU chip called the accelerated processing unit (APU).

In recent years, AMD has succeeded in taking a share from Intel of the market for CPUs used in data centers. AMD's data center revenue jumped 64% in 2023 to $6 billion, while Intel's data center sales slid 15% to $19.2 billion. AMD has made quite a few purchases in recent years to enhance its AI offerings. In 2022, AMD spent $35 billion on Xilinx, a designer of reprogrammable chips that can be made adept at AI computation after production. It is investing heavily to hire software engineers and recently acquired Silo AI for $665 million, adding 300 AI engineers to its workforce. AMD has targeted NVIDIA's H100 with its MI300X superchip designed for data centers and generative AI. AMD's data center chip combines a CPU and a GPU with memory, all in the same packaging.

A significant weakness of both AMD and Intel has been their limited software support for their AI chips. Both companies have made drivers and libraries open source as a competitive move. ROCm is AMD's open-source software stack for GPU programming, which is believed to be far behind NVIDIA's CUDA in critical aspects like documentation, performance, and adoption.

With NVIDIA struggling to meet demand for its most powerful AI chips, AMD expects to win some business. "There will certainly be scenarios in 2024 when we would imagine NVIDIA GPUs are sold out and customers only have access to AMD, and AMD can win some business that way, just based on availability,"[22] AMD CEO Su says. AMD is projecting $4 billion in data center revenues in 2024.

22 Meyer, D. (2023, October 4). AMD's Lisa Su is ready to crash Nvidia's trillion-dollar chips party. *Fortune*. https://fortune.com/2023/10/04/amd-lisa-su-chips-ai-nvidia-trillion-dollar-valuation/

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

Startups

Semiconductors are expensive to design, develop, and manufacture, making them a tough area for startups. However, venture capitalists are seeing ample opportunities for differentiation and are funding innovative companies to jump into the game. Venture capital–backed chip startups raised nearly $5.7 billion in 210 deals in the first half of 2024. In 2023, $8.8 billion was raised in 490 deals, and in 2022, $10.9 billion was raised in 447 deals.[23] There are some who say that NVIDIA's chips originally weren't even built for AI. Startups like Cerebras ($4 billion valuation), Groq (2 billion valuation), and SambaNova ($5.1 billion valuation) claim to be building chips specially designed for AI workloads.

"Nobody who started with a clean sheet of paper chose to make a GPU for this kind of work," says Andrew Feldman, CEO of Cerebras.[24] Cerebras is combining its AI hardware with the development of its own large language models (LLMs) and released Cerebras-GPT in March 2023. Cerebras's CS-3 AI accelerator chip is the size of a steering wheel and has 900,000 GPU cores with a total of 10,000,000,000,000 transistors, a staggering 4,707% increase over NVIDIA's top-of-the-line Blackwell GB200, which features 208,000,000,000.[25]

Groq started selling its chips in 2022. It offers dedicated language processing units (LPUs) with an inference engine. In a 2024 pitch deck to investors, Groq touted its LPUs as four times faster, five times cheaper, and three times more energy-efficient than NVIDIA's GPUs when used for inference. In March 2024, Groq launched GroqCloud, through which developers can rent access to its chips without buying them outright. To lure developers, Groq offered free access. In its first month, 70,000 signed up. Now, there are 280,000-plus.

Huawei

Aided by billions of dollars in Chinese government support, Huawei is a vital part of China's endeavor to overcome US sanctions and replace US technologies with homegrown alternatives. In 2023, the US government placed sanctions on NVIDIA, restricting the sale of its top-of-the-line hardware to China. Huawei's cutting-edge chip Ascend 910C rivals NVIDIA's H100 and began shipping in 2024. Alibaba, Bytedance, and Baidu, who have been key NVIDIA customers, have started purchasing chips from Huawei.

---

23 Metinko, C. (2024, June 20). Semiconductor startup funding looks to bounce back after lackluster 2023. *Crunchbase News*. https://news.crunchbase.com/semiconductors-and-5g/chip-startup-funding-bounces-back-ai-nvda/

24 Nieva, R. (2024, August 5). The AI chip boom saved this tiny startup. Now worth $2.8 billion, it's taking on Nvidia. *Forbes*. https://www.forbes.com/sites/richardnieva/2024/08/05/groq-funding-series-d-nvidia/

25 McLoughlin, A. (2024, May 20). Forget Intel and AMD—Nvidia's next big competitor might be a company you've never heard of. *TechRadar*. https://www.techradar.com/computing/gpu/forget-intel-and-amd-nvidias-next-big-competitor-might-be-a-company-youve-never-heard-of

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

The Competition That Never Was

Big tech companies like Amazon, Meta Platforms, Microsoft, and Alphabet are among NVIDIA's biggest customers, making up roughly 40% of its revenues. In 2023, Microsoft accounted for 15%, Meta for 13%, Amazon for 6%, and Alphabet for another 6% of NVIDIA's sales.[26] Amazon, Microsoft, and Alphabet are also key channel partners for NVIDIA, offering enterprises access to NVIDIA hardware through their hyperscale data centers. To limit their reliance on costly NVIDIA chips—the so-called NVIDIA tax—all four have now started developing their own in-house AI chips. They are fast turning from NVIDIA customers into competitors.

Microsoft

Microsoft is one of NVIDIA's largest customers and channel partners. It is now building its own custom AI chip that can train LLMs and potentially avoid a costly reliance on NVIDIA. Microsoft's Azure AI Accelerator (MAIA) chip and its Azure Cobalt CPU will arrive in 2024. Both custom silicon chips are designed to power Microsoft's Azure data centers and prepare the company and its enterprise customers for a future full of AI.

Microsoft's head of AI Hardware says, "Software is our core strength, but frankly, we are a systems company. At Microsoft, we are co-designing and optimizing hardware and software together so that one plus one is greater than two."[27] Much like building a house lets you control every design choice and detail, Microsoft sees the addition of homegrown chips as a way to tailor its cloud and AI workloads to itself and its customers.

Amazon

AWS, one of NVIDIA's most valuable customers and partners, has been a leader in public cloud infrastructure services, pioneering the category in 2006. It controls roughly a third of the cloud computing market and is Amazon's biggest and most profitable part, making up 70% of its total revenues.

AWS forayed into in-house chips after buying Annapurna Labs in 2015 with the design and development of its Graviton CPU chip, delivering 40% better price performance than Intel's comparable x86 CPU, for use in its data centers. In 2018, AWS introduced its first AI chip, Inferentia, designed to deliver low-cost inference in the cloud. In 2020, AWS debuted its Trainium chip to train 100-billion-plus parameter AI models. Alongside, AWS released Neuron SDK, a software development kit for developers to program Inferentia and Trainium AI chips. Customers

---

26 Bowman, J. (2024, May 2). Who are Nvidia's largest customers? *Yahoo Finance*. https://finance.yahoo.com/news/nvidias-largest-customers-201900375.html

27 Siegel, J. (2023, November 15). *With a systems approach to chips, Microsoft aims to tailor everything "from silicon to service" to meet AI demand.* Microsoft Source. https://news.microsoft.com/source/features/ai/in-house-chips-silicon-to-service-to-meet-ai-demand/

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

can't buy these chips, but they can rent them through AWS, which markets its chips as more cost-efficient than NVIDIA's.

Training AI models on Amazon chips can be half as expensive as running the same model with NVIDIA. ByteDance, the Chinese owner of TikTok, reported saving as much as 60% on the cost of deploying its AI models on AWS using Amazon's AI chips. In 2023, Amazon invested $4 billion in Anthropic, OpenAI's primary rival in the LLM space. The deal required Anthropic to use AWS as its primary cloud provider and use Amazon's custom chips to build, train, and deploy its AI models.[28]

With the largest cloud install base, Amazon has continued to build on its strengths in public cloud infrastructure services, positioning itself to become a supermarket for cloud-based AI services for developers and enterprises training AI models and integrating AI into their own applications. Many AWS AI services are available with pay-as-you-go pricing and accessible from within the AWS dashboard that customers already use to manage their compute spend. Its new managed service Amazon Bedrock, launched in 2023, offers users the ability to build AI applications on the choice of foundational model from a menu of offerings, including Amazon's in-house foundation model Titan, Anthropic's Claude, Meta's Llama, and other open-source and closed-source LLMs.

Meta

Meta isn't a cloud provider (yet). Still, the company needs massive amounts of computing power to run its software, websites, and recommendation engines that serve Facebook, Instagram, and WhatsApp ads. Meta has been spending billions on buying NVIDIA chips to power its enterprise data centers. CEO Mark Zuckerberg said the company planned to acquire roughly 350,000 flagship H100 chips from NVIDIA in 2024. However, it is also investing in building its own custom-designed data center chips to address the swelling amount of computing power necessary to run its AI applications. The Meta Training and Inference Accelerator (MTIA) chip, introduced in 2023, is being built by TSMC.

Beyond building the chips, Meta is making significant investments in developing software to harness the power of its AI infrastructure.[29] It strategically supports open-source software initiatives that seem to be aimed at breaking NVIDIA's stronghold on AI, driven by its closed ecosystem that tightly couples developers using CUDA to NVIDIA's hardware. In 2009, Meta seeded PyTorch, an open-source computing package with pre-built developer tools to set up and train deep-learning neural networks. In Feb 2023, Meta released the LLaMA family of LLMs covering a range of sizes (7 billion, 13 billion, 33 billion, and 65 billion parameters). Designed to

28 Hadero, H. (2024, March 27). Amazon pours an additional $2.75 billion into AI startup Anthropic. *AP News*. https://apnews.com/article/amazon-anthropic-investment-ai-big-tech-f5108beaa33455f331010489e03586d4

29 Cherney, M. A. (2024, April 10). Meta debuts new generation of AI chip. *Reuters*. https://www.reuters.com/technology/meta-debuts-new-generation-ai-chip-2024-04-10/

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

be proprietary to run locally but leaked and widely available on the internet as open access, the platform benefited immensely from the rapid innovation that open source makes possible. In a matter of days, various implementations emerged. For example, when model weights were changed from floating point numbers needing lots of memory to a small set of integers, the model could run even on smartphones and cost relatively little.

Google

Google has been committed to AI development since at least 2011, when it started its AI research division, Google Brain, in collaboration with Stanford University. In 2015, Google introduced its own chips (i.e., TPUs) to train and deploy AI models for in-house use in products like Google Translate and Google Maps. In 2016, it began offering developers access to its TPU chip with its free open-source software platform TensorFlow. Google's TensorFlow is a globally recognized AI framework that can run on various AI hardware, including its TPUs, AMD and Intel CPUs, and NVIDIA GPUs. In May 2024, Google announced the sixth version of its TPU, TP6, which was used to develop its AI models, including Gemini and Imagen. It also announced a new CPU Axion chip designed for data center workloads, which is expected to be released in 2024.

Google does not sell TPUs as standalone hardware products. They are offered as a cloud service (Cloud TPU) on Google Cloud, where customers can rent access to TPUs for their AI workloads. It also offers access to NVIDIA chips through its cloud. By promoting TPUs, Google reduces its dependency on NVIDIA and saves costs while offering customers a competitive, in-house option for AI workloads. While NVIDIA sells its A100 chips for about $10,000 each, Google spends an average on its own chips of just $2,000 to $3,000.

NVIDIA's AI Foundry

While the big tech companies are moving into NVIDIA's business, NVIDIA has started moving into theirs. In 2023, NVIDIA built its own cloud infrastructure, NVIDIA DGX Cloud, and began offering a new service named the NVIDIA Foundry. This is an end-to-end AI training-as-a-service for creating custom generative AI models. The service includes access to NVIDIA's AI Foundations, a collection of about 600 or so customizable pre-trained AI models in various domains that businesses can customize and fine-tune right from any browser by using NVIDIA's NeMo platform and NIM microservices.[30]

NVIDIA offers enterprise customers NVIDIA DGX Cloud and NVIDIA Foundry service directly and also through all major cloud service providers. Analysts say that NVIDIA's sheer dominance in AI

30 NIMs stands for NVIDIA Inference Microservices, and NeMo stands for Neural Modules. NVIDIA's NeMo is a framework for building and customizing AI models, while NIM is a tool for deploying and managing those models in production environments. Together, they form a powerful solution for enterprises interested in leveraging generative AI to solve real-world problems.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

is forcing cloud providers to assign dedicated data center space to the firm, which puts its DGX Cloud servers and CUDA software stack in those spaces. NVIDIA is also funneling chips and its services into a new wave of cloud startups, such as CoreWeave, that compete with the big three cloud service providers: Amazon, Google, and Microsoft.

CEO Huang explains,

> *"Generative AI is driving a platform transition and starting a new industrial revolution where AI factories will manufacture intelligence. We are an AI foundry like TSMC is a chip foundry. We manufacture custom NIMs like TSMC manufactures custom chips...In the last industrial revolution, power generation plants produced electricity. In this new industrial revolution, NVIDIA AI supercomputers are the AI factories that produce intelligence."*[31]

As NVIDIA's 2024 Annual Report notes, "Enterprises have not historically been a market for GPUs and high-performance computing. Generative AI changes that. Copilots, assistants, and agents will augment and amplify every employee and business process in the future, and enterprises will be significant markets for NVIDIA GPU computing."[32] NVIDIA's AI Foundry service centers on the idea that many companies will want to build and operate proprietary AI models, and every company will want to integrate AI into its business processes to accelerate and improve productivity. Huang says that NVIDIA is leading the AI revolution with an aim to help companies apply AI firsthand and create new markets.

Yet, There Still Are Vulnerabilities

There is no doubt that NVIDIA is at the center of the generative AI boom, and analysts predict that it will likely remain there for several years. To quote CNBC, "The sum of all that NVIDIA's doing will indeed create the next industrial revolution."[33] Yet, there are vulnerabilities.

One significant threat to NVIDIA's AI data center business may be a change in where processing happens. Developers are increasingly betting that AI work will move from data centers to the laptops, PCs, and phones we own. They think Mobile AI is coming soon. Big models like the ones developed by OpenAI require massive clusters of powerful GPUs for inference. Still, companies like Apple and Microsoft are developing small models that require less power and data, and can run on a battery-powered device. Analysts also expect a market for Edge AI—on-device processing of AI apps—to emerge. While training AI models is currently the biggest market for

---

31 NVIDIA Corporation. (2024). *2024 NVIDIA Corporation annual review: Notice of annual meeting, proxy statement, Form 10-K*. https://s201.q4cdn.com/141608511/files/doc_financials/2024/ar/NVIDIA-2024-Annual-Report.pdf

32 NVIDIA Corporation. (2024). *2024 NVIDIA Corporation annual review*. https://s201.q4cdn.com/141608511/files/doc_financials/2024/ar/NVIDIA-2024-Annual-Report.pdf

33 Coleman, J. (2024, March 18). Jim Cramer says Nvidia will "create the next industrial revolution." *CNBC*. https://www.cnbc.com/2024/03/18/jim-cramer-says-nvidia-will-create-the-next-industrial-revolution.htmll

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

chipmakers like NVIDIA, analysts say that the market will eventually shift to inferencing or running AI applications rather than training AI models. The inference process requires significantly fewer computing resources than training on a per-usage basis but will consume much more overall computing, as multiple instances of one trained AI will be deployed to do the same job in many places. For example, Microsoft's Bing AI chatbot (based on GPT-4) only had to be trained a few times, but millions of people use it daily. To address the vulnerability, NVIDIA released a compact AI supercomputer in Dec 2024, the Jetson Orin Nano Super. Priced at $249, it fits in the palm of a hand and promises gains in generative AI inference performance.

Another big challenge is that NVIDIA is competing against its biggest customers and partners. Forty percent of NVIDIA's net sales are driven by Google, Microsoft, Meta, and Amazon, which are all building their own chips to reduce their purchases from NVIDIA. The enormous size of the AI computing market has encouraged these and various other companies to unite to take on NVIDIA. Realizing that software is the main roadblock to broader adoption of the competitor chips, much of this collaboration is focused on developing open-source alternatives to CUDA. For example, a big industry group called the UXL foundation, which includes Intel, AMD, Qualcomm, Google, ARM, Samsung, and others, is working to create free open alternatives to NVIDIA's CUDA for AI applications.

Moreover, the high price of NVIDIA's chips alone is a powerful motivator for customers to seek alternatives. It also gives customers plenty of room to compete. NVIDIA's newest GPU chip costs roughly $30,000 or more. Its new machines, which include many GPUs, sell for $250,000 or more. A new supercomputer that NVIDIA is bringing to the market could sell for more than $1 million.

Up to now, NVIDIA has been winning the AI race. But being on top has its price. NVIDIA's dominance in AI has complicated its plans to acquire other companies that could complement its business. When it offered to buy ARM Holdings, which licenses designs used to create chips, for $40 billion in 2020, competitors and customers complained that the move would give NVIDIA too much power over the industry, and the deal fell through. Now, NVIDIA has offered $700 million to acquire Run.ai, a startup that controls the scheduling of AI systems. However, the US Justice Department plans to review the deal. NVIDIA's strong lead in the most advanced AI chips has also made the company a prime target of restrictions by the US government looking to slow China's AI advancement. As a result, Chinese tech giants are shifting orders to domestic chipmakers and developing their own AI hardware, significantly impacting NVIDIA's revenues.

And then there is DeepSeek, what was heretofore a little-known AI lab out of China. It ignited public panic in Jan 2025 after releasing its free, open-source AI models, V3 and R1, challenging overnight the underlying economics of training and deploying Generative AI models. DeepSeek's AI models performed as well if not better than others in performance benchmarks, despite being trained on NVIDIA's export-restrictions-compliant H800s, which were not the most advanced. Yet, according to DeepSeek it only took several months with a cost of less than $5.6 million to achieve its AI models, compared to an estimated cost of over $100 million for training OpenAI's ChatGPT-4. What if data centers didn't need to spend as much as expected on advanced NVIDIA

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

chips to develop advanced AI capability? NVIDIA stock lost $600 billion in market value in just one day.

How should NVIDIA address these vulnerabilities, and where should it go from here?

Is its growth and profitability sustainable? What must it do to sustain its dominance in the AI space? The following interactive class exercise explores these and other questions about NVIDIA's current and future strategy.

Polling Questions for Interactive Class Exercise

Participants should prepare to answer the following questions for interactive class exercise and discussion. The exercise comprises four sections, with fifteen questions in all. Participants will join the interactive exercise in the class by using their phones to scan a QR code provided by the instructor.

Section 1. Market-Creating Blue Ocean Strategy

Questions 1–3 relate to the fundamental concepts of market-creating strategy. **Select one or more answers per question.**

1. . NVIDIA's investments in building specialized AI chips for data centers can be viewed as:
   A. Competitive moves to protect its gaming market share.
   B. Speculative bets on unproven future technology.
   C. A market-creating strategy to generate new demand for AI acceleration.
   D. None of the above.
2. NVIDIA's focus on the development of the CUDA ecosystem can be viewed as a:
   A. Cost-leadership strategy to undercut competitors.
   B. Differentiation strategy to enhance customer value at a premium price.
   C. Value innovation strategy to create high buyer value at low cost for NVIDIA.
   D. None of the above.
3. NVIDIA's development of data-center-scale AI chips can be considered a blue ocean strategic move because it:
   A. Helps NVIDIA diversify beyond its core gaming GPU business.
   B. Allows NVIDIA to compete directly with Intel and AMD for the data center market.
   C. Drives NVIDIA's growth by creating a new market for AI data centers.
   D. Enables NVIDIA to offer lower-cost solutions than its rivals.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

Section 2. Strategic Alignment for Sustainable Growth

Questions 4–7 relate to the strategic alignment of three key strategy propositions required for sustainable growth. ***Select one or more answers per question****.*

4. What attracts customers to buy NVIDIA's data center AI chips?
   A. Software resources that simplify and expedite AI programming.
   B. Cost efficiency in general computing tasks.
   C. Supercomputer-like AI capabilities at a fraction of the cost.
   D. Energy efficiency.
5. How has NVIDIA created a profitable business model for its data center AI chips?
   A. By undercutting competitors on price.
   B. By offering AI solutions that command premium prices with high profit margins.
   C. By achieving lower-cost production through fabless operations and software emulation.
   D. By collaborating with the research community to expedite R&D at low cost.
6. What motivates the developer and research community to adopt NVIDIA's AI chips and contribute to its developer ecosystem?
   A. Industry-leading AI hardware with integrated software and development resources.
   B. Opportunities for professional development and career growth.
   C. Competitive pricing and exclusive deals.
   D. A platform to collaborate with other developers and researchers.
7. What is driving uncontested growth for NVIDIA's AI chips?
   A. AI acceleration generates high demand for its hardware and enables premium pricing.
   B. NVIDIA's high profit margins fund top talent and R&D to maintain technological superiority.
   C. NVIDIA's software ecosystem drives developer adoption and fends off competitors.
   D. All of the above.

Section 3. Portfolio Management Strategy

Questions 8–11 examine how a corporation like NVIDIA manages its portfolio of businesses over time to sustain its profitable growth by balancing its current profit/cash flow with its future growth. ***Select one or more answers per question.***

8. How would you classify NVIDIA's gaming GPU business?

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

A. As a game-changer creating unprecedented value—a pioneer.
B. As better than most other offerings in the marketplace—a migrator.
C. As a me-too business similar to other offerings in the market—a settler.
D. As a potentially innovative value not yet proven in the market.

9. How would you classify NVIDIA's AI data center solutions?
   A. As a game-changer creating unprecedented value—a value innovation.
   B. As better than most other offerings in the marketplace—a value improvement.
   C. As a me-too business similar to other offerings in the market—a value imitation.
   D. As a potentially innovative value not yet proven in the market.
10. How would you classify NVIDIA's AI Foundry service?
   A. As an established business creating unprecedented value.
   B. As better than most other offerings in the marketplace.
   C. As a me-too business similar to other offerings in the market.
   D. As an innovative value not yet proven in the market.
11. Which of the following best describes NVIDIA's overall strategic approach in the last 5 years?
   A. A focus on defending its core GPU market share.
   B. Aggressive diversification into creating new markets.
   C. A balanced approach of defending the core and creating new markets.
   D. None of the above.
12. . As of 2023, NVIDIA's primary source of revenue has shifted drastically from gaming to data centers. This shift can be attributed to:
   A. Declining revenue from the gaming segment.
   B. Brand-new demand for AI accelerators in data centers.
   C. Competitive threats from other chip makers.
   D. All of the above.

Section 4. NVIDIA's Future Strategy

Questions 13–15 are open-ended questions that relate to NVIDIA's future strategic options. ***Select any number of options and/or provide responses freely per question.***

13. . What is the biggest challenge facing NVIDIA's data center business?

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

A. Competition from semiconductor companies like AMD and Intel.
B. Ongoing global chip shortage and supply chain disruptions.
C. Regulatory scrutiny over its planned acquisitions.
D. Major tech companies developing in-house AI chips.
E. The high cost of R&D to develop new architectures.
F. Highly efficient and cost-effective AI training methods from companies like DeepSeek.
G. Something else: _____

14. . How should NVIDIA respond to major tech players (e.g., Amazon, Google, Microsoft, and Meta) who are developing their in-house AI chips?
    A. Develop exclusive contracts to deter them from building in-house solutions.
    B. Develop low-cost custom chips tailored to their specific needs.
    C. Reduce dependency by focusing on broadening the customer base.
    D. Sell design IP and consulting services to companies developing their own AI hardware.
    E. Fortify and protect its software and developer ecosystem.
    F. Something else: _____
15. NVIDIA's success in the future will depend on its ability to:
    A. Maintain its technological edge over competitors.
    B. Diversify its customer base beyond a few key segments.
    C. Effectively navigate regulatory challenges and acquisitions.
    D. Maintain a balanced portfolio of market-creating and market-competing initiatives.
    E. Stay ahead of the game by identifying and leveraging decisive, irreversible trends.
    F. Expand its software and services ecosystem.
    G. Something else: _____

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

Appendix 1

Background Note on Generative AI

Generative AI is a type of AI that creates new original content, such as text, images, audio, or video, in response to a user prompt. It works by learning patterns and structures from a large dataset and then using that knowledge to generate new content. OpenAI's ChatGPT, Google's Gemini, Anthropic's Claude (text generators), Github Copilot (code generator), DALL-E, Midjourney, Stable Diffusion (image generators), Synthesia (video generator), and SoundDraw (music generator) are all examples of generative AI applications.

At the core of any generative AI system is a *model,* an algorithm or a mathematical representation that maps input data to output data. The structure of the model is defined by its *architecture*, which determines how data flows through the underlying artificial neural network (ANN), including the number and types of layers and the connections between them. Computer scientists discovered the promise of generative AI in the mid-2010s when architectures like variational autoencoders (VAEs), convolutional neural networks (CNNs), generative adversarial networks (GANs), and diffusion were developed. In 2017, Google published a paper, "Attention Is All You Need," introducing a new Transformer architecture. Transformers leveraged a concept called self-attention to unlock long-term dependencies between words and understand relationships between words, even those that were far apart in a sentence. This set off a boom in generative AI development, with many large language models (*LLMs*) built on Transformer architecture. These systems, like OpenAI's GPT (Generative Pre-trained Transformer), are known as *foundation models*, where one company pre-trains a model for others to adapt and use for various downstream applications.

A generative AI model learns using a corpus of *training data***,** a collection of examples relevant to the task the model is expected to perform. The size of training data is usually expressed in terms of *tokens*, the smallest unit of data that the model can process. In a language model, for example, tokens might be individual words or subwords. GPT-3, for instance, was trained on 500 billion tokens (about 300 billion words) that comprised databases, web page metadata extracts, Reddit posts, internet-based books, and the English version of Wikipedia*. Parameters* are internal variables within the AI model that describe the linkage between data tokens. During the training process, the model learns to adjust parameter weights assumed by the architecture to improve its prediction quality with many iterations over time. After training, these parameters allow the model to make predictions when presented with new data. The number of parameters describes the size of the AI model, while the number of tokens determines the size of the dataset on which it is trained. Training a Generative AI model requires specialized hardware, like NVIDIA's GPUs in data centers, to handle heavy and massively parallel computational loads that require enormous amounts of electricity. As the model size and dataset size increase, so does the overall performance of the AI models, but also so does the computing power required and the cost.

Pre-training refers to the process of teaching models to recognize patterns and extract general features from large datasets. The process is similar to a student mastering the basics of reading, writing, arithmetic, and information processing during K–12 education before going on to specialize in a particular field during post-secondary education. Pre-training is generally *unsupervised*, meaning that instead of starting with the correct answers and working backward to arrive at a set of classification rules, the model is turned loose on the corpus of data, and the rules emerge from the clusters it abstracts from the dataset itself. After pre-training, a foundation model can be *fine-tuned* for a specific purpose using data relevant to the task. For a model pre-trained to understand the English language, for example, this might consist of focused training on expert medical knowledge to perform as a medical chatbot. Extensive fine-tuning on

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

various data types is needed if the model is expected to perform multiple tasks across different domains (e.g., a medical chatbot that can also make small talk about the weather and sports). Models are generally fine-tuned in a *supervised* manner, meaning the input data is labeled, and humans evaluate the quality of the model's output (at least initially).

After training comes the installation of *guardrails* and content policies to keep the model from generating undesired content. From there the model can be used for *inference* by developers or users through an *application programming interface* (API). Inference refers to the process of using a trained model to make predictions on unseen data. An API is a set of protocols that exposes the functionalities of a certain software to outside calls in a limited way to enable communication with other applications. When you type a query in ChatGPT, for example, the browser sends that data to a server using an API provided by OpenAI. The server receives the data, processes it through the ChatGPT model, and generates a response, which is sent back to the browser, where the output appears on the user's screen as though it is being typed back one word at a time.

The model is loaded in GPUs and fed input data for both training and inference. Traditional information retrieval models used in Google search, for example, simply search for the most relevant pages from their database, which is not a generative process. On the other hand, an LLM generates different outputs token by token to various user inputs. Given the large size of AI models and datasets, they require many GPUs running simultaneously, making it very expensive to deploy and maintain AI models.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

Appendix 2

Background Note on the Global Semiconductor Industry

A semiconductor *integrated chip (IC)* is essentially an ultra-miniaturized electronic circuit etched into a 10–20mm square silicon die capable of performing highly complex computations at light speed. The sophistication of a semiconductor IC is measured in terms of the width of the circuit traces on the silicon die, known in the industry as a process node. The narrower the node, the more transistors can be etched and the faster and more complex the calculations a circuit can perform. Nodes can vary from 600nm (nanometers) for something like an alarm clock application to around 40–28nm for cars, all the way to 5nm for the latest smartphones.

In the 1970s, semiconductor manufacturing was a vertically integrated affair: Intel Corporation, Texas Instruments Inc., and Motorola Inc. were integrated device manufacturers (IDMs) that designed, fabricated, packaged, and marketed their own ICs. As cost pressures and market size grew in the 1980s, semiconductor firms began to adopt a fabless (fabrication-less) business model, focusing on development and design while outsourcing the less profitable, capital-intensive activities to pure-play foundries focusing solely on fabrication, dominated by TSMC (Taiwan), Samsung (South Korea), and GlobalFoundries (United States) that together now account for over 75% of the global market share.

By the early 2000s, fabless design houses like NVIDIA, Broadcom, Qualcomm, and Cirrus Logic had emerged, enjoying huge margins without being saddled with the steep capital investments required for fabrication. As of early 2022, only Intel, Texas Instruments, Samsung, and a handful of other IDMs have the capability for both design and fabrication. A new 45nm process plant requires between $3bn and $4bn in capital, compared to an estimated cost of $20 billion+ for a 5nm plant, making it difficult for any new players to enter advanced chip manufacturing.

TSMC has a near monopoly over the most advanced AI chips being built today. Its latest plant in Taiwan is equipped for mass production at 3nm process nodes, a capability that no other foundry in the world can match. Samsung lags behind TSMC with the capability of mass-producing 5nm nodes since 2020. Global Foundries only has fabrication capabilities for 12nm and above. Once considered the world's most advanced chip maker, Intel was slow to master the latest fabrication processes and publicly admitted that it was two to three generations behind TSMC. It recently decided to invest $20bn in two new 7nm process fabrication plants in Arizona. TSMC's 3nm supply is now reserved by the top four tech giants: Apple, Qualcomm, NVIDIA, and AMD. With that, orders are reported to be fulfilled up to 2026. [34]

Fabricating sub-10nm process node IC chips requires extreme ultraviolet (EUV) lithography machines that are only manufactured by the Dutch company Advanced Semiconductor Materials Lithography (ASML), which has a 100% monopoly on the market.[35] Before EUV, chip fabricators could get deep ultraviolet (DUV) technology from three companies: ASML, Nikon, and Canon. ASML's latest high-numerical-aperture EUV (High-NA EUV) machines costing $300 million-plus are designed to create 2nm process node chips. Intel and TSMC are both buying ASML's High NA EUV machines in 2024.

---

34 Zuhair, M. (2024, June 13). TSMC's 3nm supply runs short amid huge AI demand: NVIDIA, Apple, AMD, Qualcomm consider raising chip prices. *Wccftech*. https://wccftech.com/tsmc-3nm-supply-runs-short-amind-huge-ai-demand-nvidia-apple-amd-qualcomm-raising-chip-prices/

35 TSMC owns 40% of ASML. ASML has exclusive deals with 800 suppliers. It would take a decade for anyone else to catch up.

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.

For the exclusive use of C. Su, 2026.

In recent years, tensions have escalated between China and the United States in a global power struggle to control the most powerful AI, with TSMC's Taiwan foundries at the center of this emerging cold war and the perceived threat of China invading Taiwan to control the AI supply chain. Restrictions enacted by the US government limit China's access to advanced technologies from ASML and TSMC, as well as the most powerful chips from NVIDIA. China's partially state-owned Semiconductor Manufacturing International Corporation (SMIC) currently has only a small share of the global fabrication market but has been expanding aggressively

Copyright © INSEAD

This document is authorized for use only by Chuanzhou Su in ECOM7121 taught by Prof. Amy Shuen, The University of Hong Kong (HKU) from Oct 2026 to Apr 2027.